



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

FULL SYLLABUS

06-04-2024

JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

ANSWER KEY

PAPER (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,C	A,C,D	B,D	A,B,D	A,C,D	C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	C	B	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	4.00	1.46	0.19	4.00	2.10 to 2.14	3.13 to 3.17	1.60	0.75

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C,D	C,D	B,C	A,B,C	A,B,D	B,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	D	A	A	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.18 to 0.22	8.00	3.00	3.50	18.00	19.00	0.00	1.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C,D	A,B	A,C,D	A,C,D	A,B,C,D	A,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	B	C	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	87.00	17.12 or 17.13	3.00	2.23 or 2.24	2.14	7.00	16.00	4.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,C)

Let the impulse due to the rod be T.

$mv - T\sin\alpha = 3mv_1$ where, V_1 is the speed of (A + C) just after the collision.

$T\cos\alpha = 3mV_2$ where V_2 is the speed of B from the inextensibility of rod AB

$$V_1\sin\alpha = V_2\cos\alpha$$

$$\text{Thus, } V_1 = \frac{v\cos^2\alpha}{3} \text{ \& } V_2 = \frac{v\sin^2\alpha}{6}$$

2. Ans (A,C,D)

$$\text{Centrifugal force} = \frac{dU}{dr} = kr = \frac{mv^2}{r},$$

$$\text{Angular momentum, } L = mvr = \frac{nh}{2\pi}$$

On solving, we get

$$r = \sqrt{\frac{nh}{2\pi\sqrt{km}}} \text{ \& } v = \sqrt{\frac{nh}{2\pi m}} \sqrt{\frac{k}{m}};$$

$$KE = \frac{1}{2}mv^2 = \frac{nh}{4\pi} \sqrt{\frac{k}{m}}$$

$$U = \frac{kr^2}{2} = \frac{k}{2} \left(\frac{nh}{2\pi\sqrt{km}} \right)$$

$$= \frac{nh}{4\pi} \sqrt{\frac{k}{m}};$$

$$E = KE + U$$

$$= \frac{nh}{2\pi} \sqrt{\frac{k}{m}}$$

3. Ans (B,D)

$$\text{Flux through frame} = \frac{\mu_0 i}{2\pi} \int_2^4 \frac{4dx}{x}$$

$$\therefore \phi = \left(\frac{2\mu_0 \ell \ln 2}{\pi} \right) i$$

$$\therefore \varepsilon = \left| \frac{d\phi}{dt} \right| = \left(\frac{2\mu_0 \ell \ln 2}{\pi} \right) \frac{di}{dt}$$

$$= \frac{2\mu_0 i_0 \omega \ell \ln 2}{\pi} \cos\left(\omega t + \frac{\pi}{6}\right)$$

$$\Rightarrow Q = \xi_c = \left(\frac{2\mu_0 i_0 \omega \ell \ln 2}{\pi} \right) \cos\left(\omega t + \frac{\pi}{6}\right)$$

$$\therefore Q_{\max} = \varepsilon_{\max} C = \left(\frac{2\mu_0 i_0 \omega \ell \ln 2}{\pi} \right) C$$

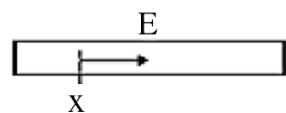
and

$$i_{\text{frame}} = \left| \frac{dq}{dt} \right| = \left(\frac{2\mu_0 i_0 \omega^2 \ell \ln 2}{\pi} \right) \left(\omega t + \frac{\pi}{6} \right)$$

4. Ans (A,B,D)

Let at any point E be field & intensity. Then

Ohm's Law



$$E = P \cdot J = \frac{P_0}{1+x} \cdot \frac{1}{\text{area}}$$

$$E = \frac{P_0}{1+x} \cdot I \quad \dots\dots\dots(i)$$

$$E \cdot dx = -dV \Rightarrow -dV = \frac{P_0 I}{1+x} \cdot dx$$

$$-\int_A^B dV = P \cdot I \int_0^1 \frac{dx}{1+x}$$

$$= P_0 I [\ln(1+x)]_0^1 = P_0 I \cdot \ln 2$$

$$V_A - V_B = (P_0 \ln 2) I \Rightarrow V = I \cdot (P_0 \ln 2)$$

$$\text{So } R_0 = \frac{V}{I} = P_0 \ln 2,$$

$$E = \frac{1}{1+x} \cdot \frac{V}{\ln 2}, E_A = \frac{V}{\ln 2} E_B = \frac{V}{2 \ln 2},$$

$$E_C = \frac{2V}{3 \ln 2}$$

5. Ans (A,C,D)

$$y = \left(\frac{\lambda \cdot D}{d} \right) \cdot n \Rightarrow n = \frac{y \cdot d}{\lambda} (D)$$

$n = 1, 2, 3, 4, \dots$ for maxima

$n = 0.5, 1.5, 2.5, \dots$ for minima

$$D = x + vt \quad n = \frac{dy}{\lambda} / [x + vt]$$

$$\text{At } t = 0, n = 4 \Rightarrow 4 = \left(\frac{yd}{\lambda} \right) / x$$

$$4x = \frac{yd}{\lambda} \Rightarrow n = \frac{4x}{x + vt},$$

$$t = \frac{x}{v}, n = \frac{4x}{2x} = 2$$

$$t = \frac{5x}{3v}, n = \frac{4x}{x + \frac{5x}{3}} = \frac{12x}{8x}$$

$$n = 1.5$$

6. Ans (C,D)

$$\frac{dA}{dt} = \frac{L}{2m}$$

$$KE_i = \frac{GMm}{2r}$$

$$U_i = \frac{GMm}{r}$$

$$TE_i = -\frac{GMm}{2r}$$

$$KE_f = \frac{6}{5} \frac{GMm}{2r}$$

$$V_f = \sqrt{\frac{6}{5} \frac{Gm}{r}}$$

MEC

$$\frac{1}{2} m \frac{6}{5} \frac{GM}{R} - \frac{Gmm}{R}$$

$$= \frac{1}{2} m v_a^2 - \frac{GMm}{r_a}$$

AMC

$$m \sqrt{\frac{6}{5} \frac{Gm}{R}} R = m v_a r_a$$

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (C)

By angular momentum conservation about the hinge,

$$L_i = L_f$$

$$\therefore mv \frac{L}{2} = \frac{mL^2}{3} \omega$$

$$\therefore \omega = \frac{3v}{2L} : t = \frac{2L}{3v}$$

$$v_{CM} = \frac{\omega L}{2} = \frac{3v}{4}$$

$$KE_f = \frac{1}{2} I \omega^2 = 25 \frac{mv^2}{L}$$

$$= \frac{1}{2} \times \frac{mL^2}{3} \times \left(\frac{3v}{2L} \right)^2$$

$$\text{Later, } F_{\text{hinge}} = m\omega^2 \frac{L}{2} = \frac{3}{8}mv^2$$

$$\text{Impulse} = m \left(v - \frac{3v}{4} \right)$$

$$= \frac{mv}{4} = m \left(\frac{3v}{2L} \right)^2 \times \frac{L}{2}$$

$$\therefore F_{\text{hinge}} = \frac{mv}{4} \div \frac{L}{100v} = \frac{9mv^2}{8L}$$

8. Ans (C)

$$F_1 = \left(\frac{a}{4} \times \rho \times g \right) \times \left(a \times \frac{a}{2} \right)$$

$$= \frac{1}{8}a^3 \rho g$$

$$F_2 = \left(\frac{3}{4}apg \right) \times \left(a \times \frac{a}{2} \right)$$

$$= \frac{3}{8}a^3 \rho g$$

$$F_3 = \left(\frac{2}{3}apg \right) \times \left(a \times \frac{a}{2} \right)$$

$$= \frac{1}{3}a^3 \rho g$$

$$F_4 = \left(\frac{1}{3}apg \right) \times \left(a \times \frac{a}{2} \right)$$

$$= \frac{1}{6}a^3 \rho g$$

9. Ans (B)

$$(i) E = \frac{KQ}{d^2} \Rightarrow E \propto \frac{1}{d^2}$$

(ii) Dipole

$$E = \frac{2kp}{d^3} \sqrt{1 + 3\cos^2\theta}$$

$$E \propto \frac{1}{d^3} \text{ for dipole}$$

(iii) For line charge

$$E = \frac{2k\lambda}{d}$$

$$E \propto \frac{1}{d}$$

$$(iv) E = \frac{2K\lambda}{d - \ell} - \frac{2K\lambda}{d + \ell}$$

$$= 2K\lambda \left[\frac{d + \ell - d + \ell}{d^2 - \ell^2} \right]$$

$$E = \frac{2K\lambda(2\ell)}{d^2 \left[1 - \frac{\ell^2}{d^2} \right]}$$

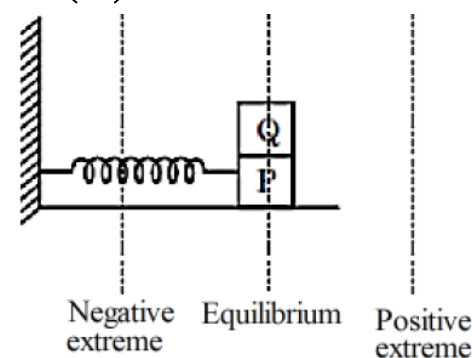
$$E \propto \frac{1}{d^2}$$

(v) Electric field due to sheet

$$E = \frac{\sigma}{2\epsilon_0}$$

$E = v$ is independent of r

10. Ans (A)



Friction on Q will always be towards equilibrium and friction on P will be opposite to that friction is only force acting on block Q in horizontal direction which is responsible for its oscillation.

Friction force will be maximum at extreme position and zero at equilibrium.

PART-1 : PHYSICS

SECTION-II

1. **Ans (4.00)**

$$\begin{aligned} \text{Area of } \triangle ABC &= \frac{1}{2} (2.5 \times 10^5) (0.05 \times 10^{-3}) \\ &= \frac{125}{2} = \frac{25}{4} \end{aligned}$$

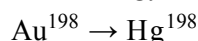
Since, $\triangle AEB$ and $\triangle BCD$ are similar, therefore

$$\frac{1.5}{2.5} = \frac{CD}{0.05} \Rightarrow CD = \frac{3}{5} \times 0.05 = 0.03$$

$$\begin{aligned} \text{Area of } \triangle BCD &= \frac{1}{2} \times (1.5 \times 10^5) (0.03 \times 10^{-3}) \\ &= \frac{4.5}{2} = \frac{9}{4} W_N = \frac{25}{4} - \frac{9}{4} = 4J \end{aligned}$$

2. **Ans (1.46)**

Total energy released from



In ground state $= (\Delta m_{\text{loss}}) c^2$

Energy released from ^{198}Hg in first

Existed state $\rightarrow \text{Hg}$ in ground state

$$= (-1.6) - (-2) \text{ MeV}$$

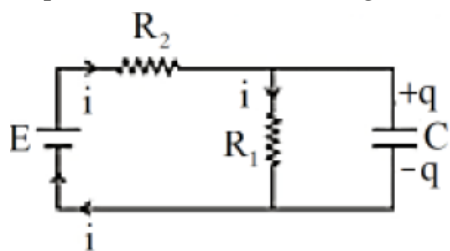
$$\Rightarrow \text{Energy released from } \text{Ag}^{198} \rightarrow \text{Hg}^{198}$$

Second excited state

$$= 1.86 - 0.4 = 1.46 = \text{max. K.E. of } \beta \text{ particle}$$

3. **Ans (0.19)**

Capacitor has maximum charge at steady state



$$\frac{q}{C} = iR_1$$

$$q = \frac{CER_1}{R_1 + R_2}$$

$$\ln q = \ln C + \ln E + \ln R_1 - \ln(R_1 + R_2)$$

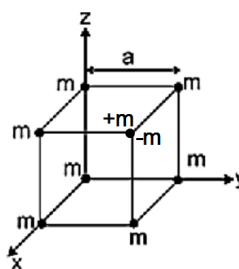
$$\frac{\Delta q}{q} = \frac{\Delta C}{C} + \frac{\Delta E}{E} + \frac{\Delta R_1}{R_1} - \left(\frac{\Delta R_1 + \Delta R_2}{R_1 + R_2} \right)$$

$$\frac{\Delta q}{q} = \frac{\Delta C}{C} + \frac{\Delta E}{E} + \Delta R_1 \left\{ \frac{1}{R_1} - \frac{1}{R_1 + R_2} \right\} - \frac{\Delta R_2}{R_1 + R_2}$$

$$\left| \frac{\Delta q}{q} \right|_{\text{max}} = \frac{0.01}{1} + \frac{4}{40} + 3 \left\{ \frac{1}{60} - \frac{1}{90} \right\} + \frac{6}{90}$$

4. **Ans (4.00)**

Place $+m$ and $-m$ particle at the vacant corner



Assume the $+m$ particle and the rest seven particle as system(1)

The total mass of system (1) is $m_1 = 8m$ and the position vector of its com is

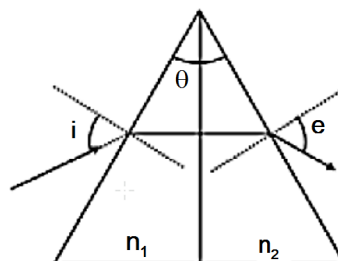
$r_1 = \frac{a}{2} \hat{i} + \frac{a}{2} \hat{j} + \frac{a}{2} \hat{k}$ and assume $-m$ particles system (ii)

$$m_2 = -m, r_2 = a\hat{i} + a\hat{j} + a\hat{k}$$

$$\begin{aligned} \vec{r}_{\text{cm}} &= \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2} \\ &= \frac{(8m) \left(\frac{a}{2} \hat{i} + \frac{a}{2} \hat{j} + \frac{a}{2} \hat{k} \right) + (-m) (a\hat{i} + a\hat{j} + a\hat{k})}{8m + (-m)} \end{aligned}$$

$$\vec{r}_{\text{cm}} = \frac{3a}{7} \hat{i} + \frac{3a}{7} \hat{j} + \frac{3a}{7} \hat{k}$$

5. **Ans (2.10 to 2.14)**



$$1 \times \sin i = \mu \sin \left(\frac{A}{2} \right),$$

$$\sin i = \frac{3}{4}$$

$$n_1 \sin 30^\circ = 1 \sin(e)$$

On differentiating both sides

$$dn \sin 30^\circ = de \cos(e)$$

$$de = \frac{dn}{2 \cos(e)}$$

$$= \frac{dn}{2 \times \frac{\sqrt{7}}{4}}$$

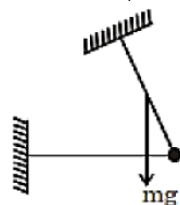
$$de = \frac{2}{\sqrt{7}} dn \Rightarrow de < dn$$

$$de = \frac{2.8 \times 10^{-3} \times 2}{\sqrt{7}} = 2.1166 \text{ mrad}$$

6. Ans (3.13 to 3.17)

$$T_1 = 2\pi \sqrt{\frac{m\ell^2/3}{mg\ell/2}} = 2\pi \sqrt{\frac{2\ell}{3g}}$$

$$T_1 = 2\pi \sqrt{\frac{m\ell^2}{3 \left(mg\frac{\ell}{2} + k\ell^2 \right)}}$$



$$= 2\pi \sqrt{\frac{m\ell}{3 \left(\frac{3mg}{2} \right)}}$$

$$- \left(mgx\frac{\ell}{2}\theta + k\ell\theta\ell \right) = \frac{m\ell^2}{3}\alpha$$

$$= 2\pi \sqrt{\frac{2}{9} \frac{\ell}{g}}$$

$$\alpha = \frac{\left(mgx\frac{\ell}{2} + k\ell^2 \right)}{\frac{m\ell^2}{3}}\theta$$

$$T = \pi \sqrt{\frac{2\ell}{3g}} + \pi \sqrt{\frac{2\ell}{9g}}$$

$$= \pi \sqrt{\frac{2\ell}{3g}} \left(1 + \frac{1}{\sqrt{3}} \right)$$

$$= 2(1+0.577) = 3.15 \text{ sec}$$

7. Ans (1.60)

$$f_1 = f_0$$

$$f_2 = \frac{c+v}{c-v} \times f_0$$

$$f_b = 256 \left(\frac{2v}{c-v} \right) = 1.6\text{Hz}$$

8. Ans (0.75)

$$F = (\alpha y \hat{i} + 2\alpha x \hat{j})$$

$$W_{AB} = (-1\hat{i}) \cdot (1\hat{i}) = -1\text{J}$$

$$\left[\begin{array}{l} \vec{F} = -1\hat{i} + 2\alpha x\hat{j} \\ \vec{S} = 1\hat{i} \end{array} \right]$$

Similarly,

$$W_{BC} = 1\text{J}$$

$$W_{CD} = 0.25\text{J}$$

$$W_{DE} = 0.5\text{J}$$

$$W_{EF} = W_{FA} = 0\text{J}$$

$$\therefore \text{New work in cycle} = 0.75 \text{ J}$$

PART-2 : CHEMISTRY

SECTION-I (i)

3. Ans (B,C)

B) Reactivity towards water $\uparrow$ from Li to Cs due to the $\downarrow$ melting points of metals

C) fact

D) KCl solubility in $\text{H}_2\text{O} = 27.77 \text{ g/100 mL } (0^\circ\text{C})$

KCl solubility in $\text{H}_2\text{O} = 36 \text{ g/100 ml pure water at}$

$T = 25^\circ\text{C}$

5. Ans (A,B,D)

10 mole A and 10 mole B

$$P_T = P_A^0 X_A + P_B^0 X_B$$

$$P_T = 0.6 \times \frac{1}{2} + 9 \times \frac{1}{2}$$

$$P_T = 0.75$$

Since $1/4^{\text{th}}$ mole will be converted

$\therefore$ In vapour 4 moles total will be present.

Assuming x mole to be of A.

	Liquid	Vapour
A	$10 - x$	x
B	$5 + x$	$5 - x$
(i)	$\frac{x}{5} = \frac{10-x}{15} \times \frac{0.6}{P_T}$	
	$P_T = \frac{(10-x) \times 0.6}{3x}$	
(ii)	$\frac{5-x}{5} = \frac{5+x}{15} \times \frac{0.9}{P_T}$	
	$P_T = \frac{(5-x)}{3(5-x)} \times 0.9$	
	$\Rightarrow \frac{(10-x) \times 6}{3x} = \frac{(5+x) \times 9}{3(5-x)}$	

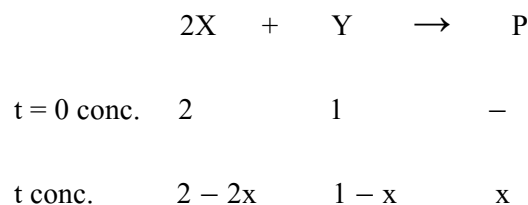
Simplifying $(x^2 + 45x - 100) = 0$

$$\Rightarrow x = \frac{-45 + \sqrt{2425}}{2} = 2.125$$

PART-2 : CHEMISTRY

SECTION-I (ii)

8. Ans (A)



$$t = 50 \text{ sec } n_Y = 0.5 \Rightarrow t_{1/2} = 50 \text{ sec}$$

$$A) 2K_{rxn}t = \ln \left(\frac{X_0}{X_t} \right)$$

$$t_{1/2} = \frac{\ln 2}{2K_{rxn}} = \frac{0.693}{2 \times 2_{rxn}} = 50$$

$$K_{rxn} = 0.693 \times 10^{-2} = 6.93 \times 10^{-3} \text{ sec}^{-1}$$

(A) is correct

(B) is correct

$$(C) -\frac{d[X]}{dt} = 2K_{rxn}[X]$$

$$= 2 \times 6.93 \times 10^{-3} \times 1$$

$$= 13.86 \times 10^{-3} \text{ M sec}^{-1} \text{ (correct)}$$

$$(D) -\frac{d[Y]}{dt} = K_{rxn}[X]$$

9. Ans (A)

$$(P) \underbrace{a \xrightarrow[=18 \text{ sec}]{t_{1/3}} \frac{2a}{3} \xrightarrow[=18 \text{ sec}]{t_{1/3}} \frac{4a}{9} \xrightarrow[=18 \text{ sec}]{t_{1/3}} \frac{8a}{27}}_{t_{19/17}=54 \text{ sec}}$$

$$(Q) \underbrace{a \xrightarrow[=16 \text{ sec}]{t_{1/4}} \frac{3a}{4} \xrightarrow[=16 \text{ sec}]{t_{1/4}} \frac{9a}{16}}_{t_{7/16}=32 \text{ sec}}$$

$$(R) K = \frac{a - \frac{2a}{3}}{18} = \frac{x}{30}$$

$$\Rightarrow x = \frac{5}{9}a$$

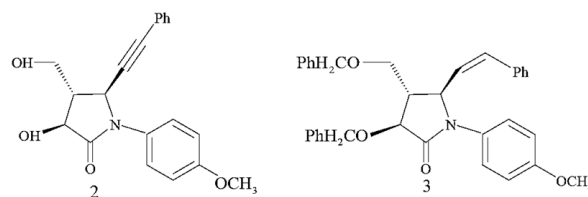
$$(S) K = \frac{a - \frac{a}{2}}{16} = \frac{x}{28}$$

$$\Rightarrow x = \frac{7}{8}a$$

PART-2 : CHEMISTRY

SECTION-II

1. Ans (0.18 to 0.22)



The molecular weight of final product is 519.

0.20 moles of reactant gives 50.6 gm of final product with the given yield.

2. Ans (8.00)

(i) IF_5 (sp^3d) (ii) ICl_4^- (sp^3d^2)

(iii) XeO_2F_2 (sp^3d) (iv) NH_2^- (sp^3)

(v) BCl_3 (sp^2) (vi) $BeCl_2$ (sp)

(vii) $AsCl_4^+$ (sp^3) (viii) $B(OH)_3$ (sp^2)

(ix) NO_2^- (sp^2) (x) ClO_2^+ (sp^2)

$$x = (sp^3)^2 + sp^3d(1) + sp^3d^2(2) = 5$$

$$y = 4, z = 1; 5 + 4 - 1 \Rightarrow 8$$

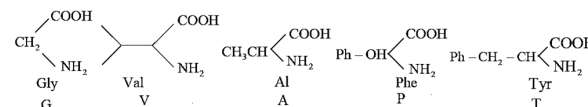
3. Ans (3.00)

$$x=0, y=3$$

4. Ans (3.50)

All give oxygen on heating.

5. Ans (18.00)

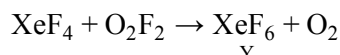


$\left. \begin{array}{l} \text{V G A P T} \quad \text{V G P A T} \\ \text{V A G P T} \quad \text{V P G A T} \\ \text{V A P G T} \quad \text{V P A G T} \end{array} \right\} \text{ 6 combinations by}$

each of V, A, P

$\therefore$ Total 18 possible sequences.

6. **Ans (19.00)**



Y has 3 lone pair of electron in each fluorine and one lone pair of electron in xenon.

Hence total lone pair of electrons is 19.

7. **Ans (0.00)**

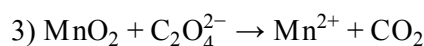
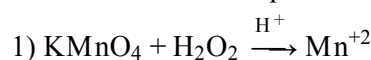
$$\Delta S_{\text{cycle}} = 0 = \frac{q_1}{T_1} + \frac{q_1}{T_2} = 0,$$

$$nR \ln \frac{V_2}{V_1} + nR \ln \frac{V_4}{V_3} = 0,$$

$$nR \ln \frac{V_2 V_4}{V_1 V_3} = 0$$

8. **Ans (1.00)**

Reaction used in the question



PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (A,B,C,D)**

$$\frac{1}{x} + \frac{1}{y} = 1$$

$$\Rightarrow x + y = xy$$

$$\Rightarrow (x-1)(y-1) = 1$$

$$\Rightarrow x = 2, y = 2$$

2. **Ans (A,B)**

$$(1-x^3)^n = \sum_{r=0}^n \frac{a_r x^r (1-x)^{3n}}{(1-x)^{2r}}$$

$$\Rightarrow \left(\frac{1-x^3}{(1-x)^3} \right)^n = \sum_{r=0}^n a_r \left(\frac{x}{(1-x)^2} \right)^r$$

$$\Rightarrow \left(\frac{1+x+x^2-3x+3x}{(1-x)^2} \right)^n = \sum_{r=0}^n a_r \left\{ \frac{x}{(1-x)^2} \right\}^r$$

$$\Rightarrow \left(1 + \frac{3x}{(1-x)^2} \right)^n = \sum_{r=0}^n a_r \left\{ \frac{x}{(1-x)^2} \right\}^r$$

$$\Rightarrow \sum_{r=0}^n {}^nC_r 3^r \left(\frac{x}{(1-x)^2} \right)^r = \sum_{r=0}^n a_r \left\{ \frac{x}{(1-x)^2} \right\}^r$$

$$\Rightarrow a_r = {}^nC_r 3^r$$

3. **Ans (A,C,D)**

$S(n, k) \equiv$ No. of ways to partition n distinct items into k parts.

First consider one particular element a_1 forming one group and remaining $n-1$.

elements forming $k-1$ partitions.

Second, consider $n-1$ elements forming k partitions and a_1 element going in one of ' k ' groups.

$$\text{Hence, } S(n, k) = S(n-1, k-1) + kS(n-1, k)$$

$$\text{No. of onto functions} = k!S(n, k)$$

$$= \left[k^{n-k} {}^nC_1 (k-1)^{n-k} {}^nC_2 (k-2)^2 \dots \right]$$

$$S(5, 3) = \frac{1}{3!} [3^{5-5} {}^C_1 2^5 + {}^C_2 3^2 1^3] = 25$$

$$\text{No. of onto functions from 6 elements to 3 elements} = 3!S(6, 3)$$

4. **Ans (A,C,D)**

Any point on the given line will be

$$(2\lambda+1, -3\lambda-1, \lambda) \equiv (\alpha, \beta, \gamma)$$

$$\therefore \alpha + \beta + \gamma = 0$$

$$\text{Now } T_{r+1} = {}^9C_r (2x^2)^{9-r} (-x)^{-r}$$

$$= (-1)^r {}^9C_r (2x^2)^{9-r} x^{18-3r}$$

$$18-3r = 0 \Rightarrow r = 6$$

$$\therefore \text{Coefficient of } x^0 = {}^9C_6 2^3$$

$$= \frac{9 \times 8 \times 7}{6} \times 8 = 84$$

5. **Ans (A,B,C,D)**

$$x\vec{a} + y\vec{b} - \vec{c} = \vec{0}$$

$$\Rightarrow \text{If } x + y - 1 = 0$$

then C lies on line segment AB.

If $x > 0$, $y > 0$ and $x + y - 1 = 0$ then C lies segment AB.

If $x < 0$, $y > 0$ and $x + y - 1 = 0$ then C lies on line but not on line segment AB.

If $x > 0$, $y < 0$ and $x + y - 1 = 0$ then C lies on line but not on line segment

If $x > 0$, $y > 0$ and $x + y - 1 < 0$ the C lie inside $\triangle AOB$

6. Ans (A, C)

$$A + B^T = B^2$$

$$B + A^T = A^2$$

$$\Rightarrow (A - A^T) - (B - B^T) = B^2 - A^2$$

$\Rightarrow B^2 - A^2$ is skew symmetric matrix of odd order

$$|B^2 - A^2| = 0$$

Now take transpose of equation $B^T = B^2 - A$

$$\Rightarrow B = (B^T)^2 - A^T$$

$$\Rightarrow (B^T)^2 - B = A^T$$

$$\Rightarrow (B^T)^2 - B = A^2 - B$$

$$\Rightarrow (B^T)^2 - B = A^2 - B$$

$$\Rightarrow (B^T)^2 = A^2$$

$$\Rightarrow |B| = \pm |A|$$

$$\Rightarrow (B^T)^2 - B = (B^2 - B^T)^2 - B$$

$$\Rightarrow B^4 - B^2 B^T - B^T B^2 = 0$$

$$\Rightarrow B^2 (B^2 - B^T) = B^T B^2$$

$$\Rightarrow |B^2| |B^2 - B^T| = |B|^3$$

$$\Rightarrow |B|^2 |A| = |B|^3, \text{ similarly } |A|^2 |B| = |A|^3$$

$\Rightarrow$ If $|B| \neq 0$ then $|A| = |B|$ and if $|B| = 0$ then $|A| = 0$

$$\Rightarrow |A| = |B|$$

PART-3 : MATHEMATICS

SECTION-I (ii)

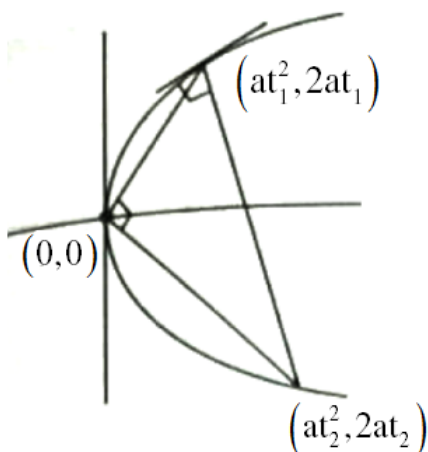
7. Ans (A)

(P) For line to become tangent to ellipse

$$a^2 = 2 + 1$$

$$\Rightarrow a = \pm\sqrt{3}$$

(Q)



$$\frac{2}{t_1} \frac{2}{t_2} = -1, t_1 t_2 = -4$$

$$t_2 = -t_1 - \frac{2}{t_1} = -\frac{4}{t_1}$$

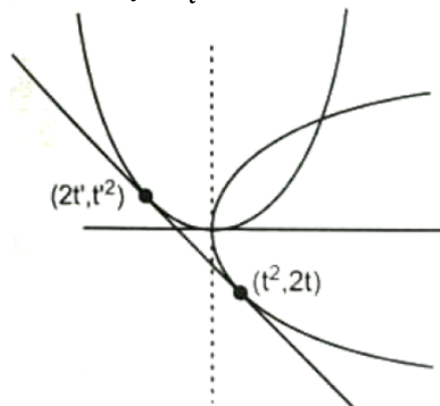
$$m = -t_1 = \pm\sqrt{2}$$

$$(R) ty = x + t^2 \Rightarrow ty - x = t^2 \quad \text{--- (1)}$$

$$t'x = y + t'^2 \Rightarrow t'x - y = t'^2 \quad \text{--- (2)}$$

(1) and (2) are identical

$$\Rightarrow -t = -\frac{1}{t} = \frac{t^2}{t^2} = t^4$$



$$t^3 = -1 \Rightarrow t = -1$$

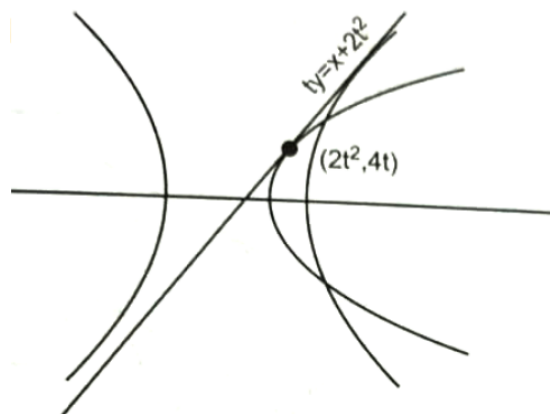
$$-y - x = 1$$

$$x + y + 1 = 0 \Rightarrow k = \sqrt{3}$$

$$(S) ty = x + 2t^2$$

$$y = \frac{1}{t}x + 2t$$

For line to become tangent to hyperbola.



$$4t^2 = \frac{1}{t^2} - 3 \Rightarrow 4t^4 + 3t^2 - 1 = 0$$

$$\Rightarrow (4t^2 - 1)(t^2 + 1) = 0$$

$$t = \pm\frac{1}{2}$$

$$\pm\frac{1}{2}y = x + \frac{1}{2}$$

$$\Rightarrow 2x + 1 = \pm y$$

$$2x \pm y + 1 = 0$$

8. Ans (B)

$$(P) P(3 \cos \theta, 2 \sin \theta) \equiv \left(\frac{3(1 - \tan^2 \theta)}{(1 + \tan^2 \frac{\theta}{2})}, \frac{4 \tan \frac{\theta}{2}}{(1 + \tan^2 \frac{\theta}{2})} \right)$$

$\tan \frac{\theta}{2}$ can take infinite rational values

$$(Q) P(\pm 3, 0), (0, \pm 2)$$

$$(R) P(\sqrt{3} \cos \theta, \sin \theta)$$

$$\frac{\tan \theta}{\sqrt{3}} = k \in \mathbb{Q}$$

$$\Rightarrow P\left(\frac{\sqrt{3}}{\sqrt{1+3k^2}}, \frac{\sqrt{3}k}{\sqrt{1+3k^2}}\right)$$

$$\text{is } \frac{\sqrt{1+3k^2}}{\sqrt{3}} = \lambda \quad \lambda \in \mathbb{Q}$$

$$\lambda^2 - k^2 = \frac{1}{3}$$

Which can be expressed as product of two rational in infinite ways say for example

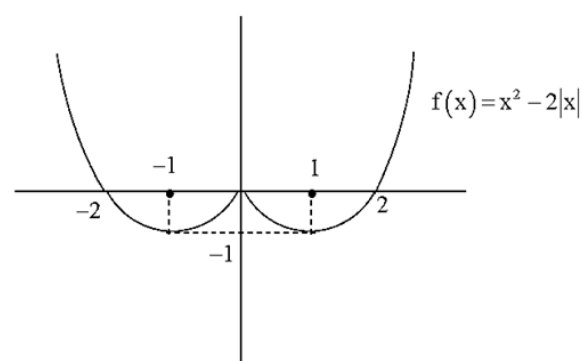
$$\lambda - k = \frac{1}{2};$$

$$\lambda + k = \frac{2}{3}$$

$$\Rightarrow \lambda = \frac{7}{12}, k = \frac{1}{12}$$

$$(S) P \equiv (0, \pm 1)$$

10. Ans (D)



$$g(x) = \begin{cases} f(x) = x^2 + 2x; & -2 \leq x \leq 1 \\ -1 & -1 < x < 0 \\ 0 & 0 \leq x \leq 2 \\ f(x) = x^2 - 2x; & 2 < x \leq 3 \end{cases}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (87.00)

If child buys pencil on the first day then the remaining $n-1$ rupees can be spend in S_{n-1} ways and if he buys pen on the first day then the remaining $n-2$ rupees can be spend in S_{n-2} ways for book on the first day.

$\therefore$ the recurrence relation is

$$S_n = S_{n-1} + S_{n-2} + S_{n-2}$$

$$\therefore S_n = S_{n-1} + 2S_{n-2}$$

$$\text{Clearly } S_1 = 1, S_2 = 3 \{(1,1), 2, 2\}$$

$$\text{Now, } S_3 = S_2 + 2S_1 = 5$$

$$S_4 = 5 + 2 \cdot 3 = 11$$

$$S_5 = 11 + 2 \cdot 5 = 21$$

$$S_6 = 21 + 2 \cdot 11 = 43$$

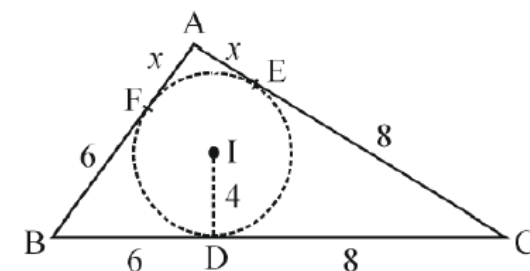
$$S_7 = 43 + 2 \cdot 21$$

$$= 43 + 42$$

$$= 85$$

$$\therefore \left(\frac{S_n - S_{n-1}}{S_{n-2}} + S_7 \right) = 85 + 2 = 87$$

2. Ans (17.12 or 17.13)



$$s = x + 14$$

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

$$\Delta = \sqrt{(x+14)(x)(8)(6)} \quad \therefore \Delta = r \times s$$

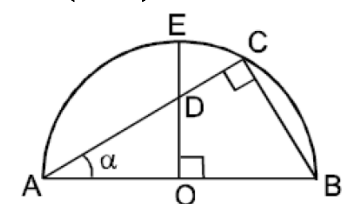
$$\sqrt{8 \times 6x(x+14)} = 4(x+14)$$

$$\Rightarrow x = 7 \Rightarrow \Delta = 84$$

$$\sqrt{\Delta - 3} = 9$$

$$R = \frac{abc}{4\Delta} = \frac{65}{8} = 8.125$$

3. Ans (3.00)



$$1 + 2 \cos \alpha - \frac{1}{\cos \alpha}$$

$$= \tan \alpha + 2 \sin \alpha$$

$$\Rightarrow \cos \alpha + \cos 2\alpha = \sin \alpha + \sin 2\alpha \left(\cos \frac{\alpha}{2} \neq 0 \right)$$

$$\Rightarrow \tan \frac{3\alpha}{2} = 1 \Rightarrow \alpha = \frac{\pi}{6}$$

4. Ans (2.23 or 2.24)

$$\text{Since } z_1 + z_2 + z_3 = 1$$

$$\Rightarrow \overline{z_1 + z_2 + z_3} = 1$$

$$\Rightarrow \frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} = 1$$

$$\Rightarrow z_2 z_3 + z_3 z_1 + z_1 z_2 = 1$$

Now, consider a cubic equation having roots z_1 ,

z_2, z_3

$$(z - z_1)(z - z_2)(z - z_3) = z^3 - (z_1 + z_2 + z_3)z^2 +$$

$$(z_1 z_2 + z_2 z_3 + z_3 z_1)z - z_1 z_2 z_3 = 0$$

$$\Rightarrow z^3 - z^2 + z - 1 = 0$$

$$\Rightarrow z^2(z - 1) + 1(z - 1) = 0$$

$$\Rightarrow z = 1, z = \pm i$$

$$\therefore \text{Roots } z_1 = -i, z_2 = 1, z_3 = i$$

$$|z_1 + z_2^2 + z_3^3| = \sqrt{5}$$

5. Ans (2.14)

$$\sum_{n=0}^{\infty} \tan^{-1} \left(\frac{\tan^{-1} \left(\frac{(n+2)-(n+1)}{1+(n+1)(n+2)} \right)}{1 + \tan^{-1} \left(\frac{1}{n+1} \right) \tan^{-1} \left(\frac{1}{n+2} \right)} \right)$$

$$\sum_{n=0}^{\infty} \tan^{-1} \left(\frac{\tan^{-1} \left(\frac{1}{n+1} \right) - \tan^{-1} \left(\frac{1}{n+2} \right)}{1 + \tan^{-1} \left(\frac{1}{n+1} \right) \cdot \tan^{-1} \left(\frac{1}{n+2} \right)} \right)$$

$$\sum_{n=0}^{\infty} \left(\tan^{-1} \left(\tan^{-1} \left(\frac{1}{n+1} \right) \right) \right) - \tan^{-1} \left(\tan^{-1} \left(\frac{1}{n+2} \right) \right)$$

$$\tan^{-1} (\tan^{-1} (1)) = \tan^{-1} \left(\frac{\pi}{4} \right)$$

$$\therefore p = \pi$$

$$\cos^{-1} (\cos(\pi - 1)) = \pi - 1 = 2.14$$

6. Ans (7.00)

$$\text{Given } f(x+5) \geq f(x) + 5 \quad \text{---(1)}$$

$$f(x+1) \leq f(x) + 1 \quad \text{---(2)}$$

From (1) and (2) we can write

$$f(x) + 5 \leq f(x+5) \leq f(x+4) + 1$$

$$\leq f(x+3) + 2$$

$$\leq f(x+2) + 3$$

$$\leq f(x+1) + 4$$

$$\leq f(x) + 5$$

$\therefore$ Equality exist every where

$$\therefore f(x+1) = f(x) + 1 \quad \because f(1) = 1$$

$$\Rightarrow f(2) = 2, f(3) = 3 \dots \dots \dots f(2023) = 2023.$$

$$\text{Now } g(x) = f(x) + 7 - x \Rightarrow g(2023) = 7$$

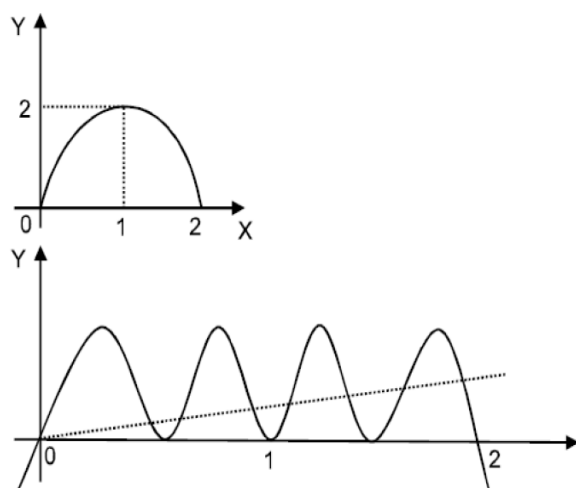
7. Ans (16.00)

$$f(x) = 2x(2 - x)$$

$$y = f(f(f(x))) ; 0 \leq x \leq 2$$

$$y = 32x(2 - x)(x - 1)^2(2x^2 - 4x + 1)^2$$

$$; 0 \leq x \leq 2$$



8. Ans (4.00)

Equation of tangent at (x_i, y_i) on the curve

$$y = f(x) \text{ is: } Y - y_i = m(X - x_i)$$

$$\text{Put } Y = 0, X = x_{i+1},$$

$$\text{then } 0 - y_i = m(x_{i+1} - x_i)$$

$$\Rightarrow x_{i+1} - x_i = -\frac{y_i}{m}$$

$$\therefore \frac{x_{i+1}}{x_i} - 1 = -\frac{y_i}{mx_i}$$

If $x_1, x_2, x_3, \dots, x_n$ form G.P.

$$\text{then } r - 1 = -\frac{y_i}{mx_i} \text{ (Where } r \text{ is common ratio of}$$

G.P.)

$$\therefore m = \frac{dy}{dx} = -\frac{y}{(r-1)x}$$

$$\Rightarrow \frac{dy}{y} + \frac{dx}{(r-1)x} = 0$$

$$\Rightarrow \ell n y + \left(\frac{1}{r-1} \right) \ell n x = k$$

$$\Rightarrow y(x)^{\frac{1}{r-1}} = c$$

$$\Rightarrow r = 2, xy = c, e = \sqrt{2}$$